Lecture 15, Mar 02 - 2026

Rationalizing Substitutions

Example. Evaluate $\int \frac{\sqrt{x+4}}{x}\,dx$.

Sln. let $u=\sqrt{x+4}$ then $x=u^2-4$. then $du=\frac{1}{2\sqrt{x+4}}dx=\frac{1}{2u}dx$

then $dx=2u\,du$.

$$I=\int \frac{\sqrt{x+4}}{x}\,dx=\int \frac{u\,(2u)\,du}{u^2-4}=2\int \frac{u^2}{u^2-4}\,du.$$

$$\begin{array}{r|l} u^2 & u^2-4 \\ \hline -u^2+4 & 1 \\ \hline 4 & \end{array}$$ then

$$I=2\int \frac{u^2}{u^2-4}\,du=2\int\left(1+\frac{4}{u^2-4}\right)du.$$

$$=2u+8\int\frac{1}{u^2-4}\,du.$$

$$=2u+8\int\frac{1}{(u+2)(u-2)}\,du$$

$$\frac{1}{(u+2)(u-2)}=\frac{A}{u+2}+\frac{B}{u-2}=\frac{A(u-2)+B(u+2)}{(u+2)(u-2)}$$

$u=2$ then $1=4B \Rightarrow B=1/4$.

$u=-2$ then $1=-4A \Rightarrow A=-\frac{1}{4}$ then

$$I=2u+8\left\{-\frac{1}{4}\int\frac{1}{u+2}\,du+\frac{1}{4}\int\frac{1}{u-2}\,du\right\}$$

$$I=2u+8\left(-\frac{1}{4}\ln|u+2|+\frac{1}{4}\ln|u-2|\right)+C$$

$$I=2u+2\ln\left|\frac{u-2}{u+2}\right|+C$$

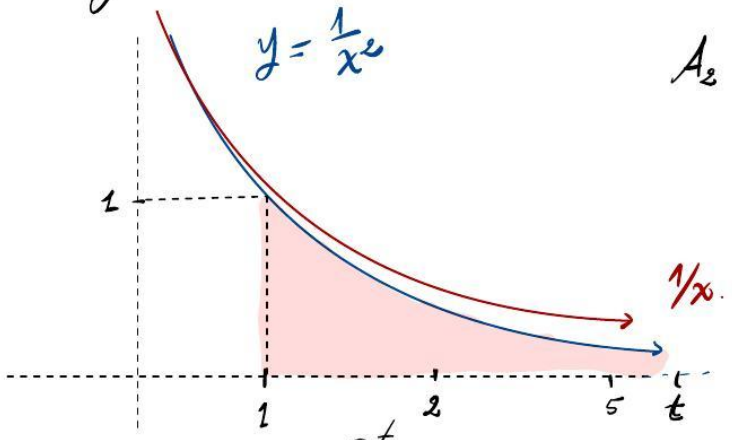
$$I=2\sqrt{x+4}+2\ln\left|\frac{\sqrt{x+4}-2}{\sqrt{x+4}+2}\right|+C.$$

7.8. Improper integrals.

Type 1. Infinite intervals.

Let consider $y = 1/x^2$, above the x-axis and to the right of $x=1$.

You might think that S is infinite



$$A_2 = \int_1^2 \frac{1}{x^2}\,dx = -\frac{1}{x}\Big|_1^2 = \frac{1}{2}.$$

$$A_5 = \int_1^5 \frac{1}{x^2}\,dx = -\frac{1}{x}\Big|_1^5 = \frac{4}{5}.$$

then $A(t) = \int_1^t \frac{1}{x^2}\,dx$ this a new function on the t variable.

then, $A(t) \xrightarrow{?} \square$ as $t \to \infty$. By definition

$$\lim_{t\to\infty} A(t) = \lim_{t\to\infty} \int_1^t \frac{1}{x^2}\,dx := \int_1^\infty \frac{1}{x^2}\,dx$$

then $$\lim_{t\to\infty} \int_1^t \frac{1}{x^2}\,dx = \lim_{t\to\infty} \left\{ -\frac{1}{x}\Big|_1^t \right\}$$

$$= \lim_{t\to\infty} \left\{ -\frac{1}{t} + 1 \right\} = 1.$$

$$\int_1^\infty \frac{1}{x}\,dx = \lim_{t\to\infty} \int_1^t \frac{1}{x}\,dx$$

$$= \lim_{t\to\infty} \left\{ \ln|x| \Big|_1^t \right\}$$

$$= \lim_{t\to\infty} \ln|t| = \infty$$

Def. a) If $\int_a^t f(x)\,dx$ exists for every number $t \geq a$ then

$$\int_a^\infty f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx.$$

b). If $\int_t^b f(x)\,dx$ exists for every number $t \le b$, then

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{t\to-\infty} \int_t^b f(x)\,dx$$

The improper integrals $\int_a^{\infty} f(x)\,dx$, $\int_{-\infty}^{b} f(x)\,dx$ are called convergent if the corresponding limit exists and divergent if the limit does not exists.

c). If both $\int_a^{\infty} f(x)\,dx$ and $\int_{-\infty}^{a} f(x)\,dx$ are convergent then

$$\int_{-\infty}^{\infty} f(x)\,dx = \underbrace{\int_{-\infty}^{a} f(x)\,dx}_{\text{case b)}} + \underbrace{\int_a^{\infty} f(x)\,dx}_{\text{case a)}}$$

Example. Evaluate $\int_{-\infty}^{0} xe^x\,dx$

Sln:

$$\int_{-\infty}^{0} xe^x\,dx = \lim_{t\to-\infty} \int_t^0 xe^x\,dx \qquad u = x \quad dv = e^x$$

$$du = 1 \quad v = e^x$$

by parts

$$= \lim_{t\to-\infty} \left\{ xe^x \Big|_t^0 - \int_t^0 e^x\,dx \right\}$$

$$= \lim_{t\to-\infty} \left\{ -te^t - e^x \Big|_t^0 \right\}$$

$$= \lim_{t\to-\infty} \left\{ -te^t - 1 + e^t \right\}$$

$$= -1 + \lim_{t\to-\infty} e^t(1-t) \qquad e^t(1-t) = \frac{(1-t)}{e^{-t}}$$

$$= -1 + \lim_{u\to\infty} \frac{1+u}{e^u}$$

$$\overset{L'H}{=} -1 + \lim_{u\to\infty} \frac{1}{e^u} = -1$$

Example: Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$

Sln. Under the assumption that $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$ is convergent then

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + \int_{0}^{\infty} \frac{1}{1+x^2}\,dx.$$

- $$\int_{-\infty}^{0} \frac{1}{1+x^2}\,dx = \lim_{t\to-\infty} \int_{t}^{0} \frac{1}{1+x^2}\,dx = \lim_{t\to-\infty} \left\{ \arctan(x) \Big|_{t}^{0} \right\}$$

$$= \lim_{t\to-\infty} \{\arctan(0) - \arctan(t)\}$$

$$= -\lim_{t\to-\infty} \arctan(t) = \frac{\pi}{2}$$

- $$\int_{0}^{\infty} \frac{1}{1+x^2}\,dx = \lim_{t\to\infty} \left\{ \arctan(x) \Big|_{0}^{t} \right\}$$

$$= \lim_{t\to\infty} \arctan(t) = \frac{\pi}{2}.$$

then $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$ is convergent and

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

Example: For what values of p is the integral

$$\int_{1}^{\infty} \frac{1}{x^p}\,dx$$

convergent?

Sln.

$$\int_{1}^{\infty} \frac{1}{x^p}\,dx = \lim_{t\to\infty} \int_{1}^{t} x^{-p}\,dx \quad , \; p \neq 1.$$

$$= \lim_{t\to\infty} \frac{x^{1-p}}{1-p} \Big|_{1}^{t}$$

$$= \lim_{t\to\infty} \left\{ \frac{t^{1-p}}{1-p} - \frac{1}{1-p} \right\}$$

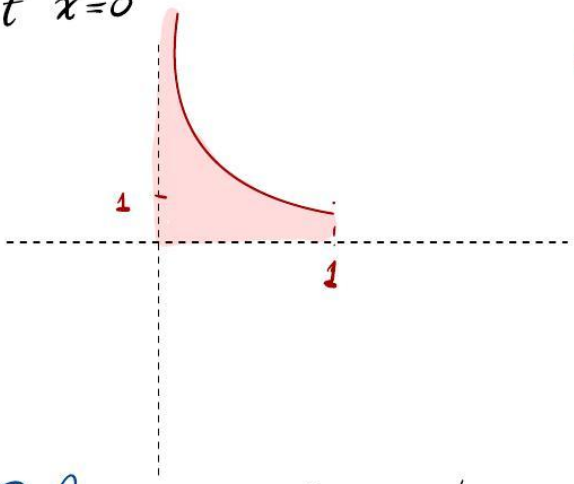
$$= \frac{1}{1-p} \lim_{t\to\infty} \left\{ \frac{1}{t^{p-1}} - 1 \right\} = \begin{cases} \frac{1}{p-1} & \text{if } p > 1. \\ \text{Divergent if } p \leq 1. \end{cases}$$

$\int_2^\infty \frac{1}{x^{0.99999}}\,dx$ this a p-Integral with $p = 0.99999 < 1$.

then is divergent.

Type 2. Discontinuous integrands

For example $\frac{1}{x}$ and $\int_0^1 \frac{1}{x}\,dx$ and $f(x) = \frac{1}{x}$ is not defined at $x = 0$



if x_0 is a point where f is not continuous then $\int_a^{x_0} f(x)\,dx = \lim_{t \to x_0^-} \int_a^t f(x)\,dx$

Def. a). If f is continuous $[a, b)$ and is discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{t \to b^-} \int_a^t f(x)\,dx.$$

if this limit exists.

b). If f is continuous on $(a, b]$ and is discontinuous at a, then

$$\int_a^b f(x)\,dx = \lim_{t \to a^+} \int_t^b f(x)\,dx.$$

if this limit exists.

The improper integral $\int_a^b f(x)\,dx$ is called convergent if the corresponding limit exists and divergent if the limit does not exist.

c). If has a discontinuity at c, where $a < c < b$ and both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ are convergent, then we define

$$\int_a^b f(x)\,dx = \underbrace{\int_a^c f(x)\,dx}_{\text{type a)}} + \underbrace{\int_c^b f(x)\,dx}_{\text{type b)}}$$

Example. find $\int_2^3 \frac{1}{\sqrt{x-2}}\,dx$

Sln. $2 \notin \text{Dom}\left(\frac{1}{\sqrt{x-2}}\right)$. then

$$\int_2^3 \frac{1}{\sqrt{x-2}}\,dx = \lim_{t\to 2^+} \int_t^3 \frac{1}{\sqrt{x-2}}\,dx$$

$$= \lim_{t\to 2^+} 2\sqrt{x-2}\Big|_t^3$$

$$= \lim_{t\to 2^+} \left\{2\sqrt{3-2} - 2\sqrt{t-2}\right\}$$

$$= 2 \lim_{t\to 2^+} \left\{1 - \sqrt{t-2}\right\} = 2.$$

Example: i). $\int_0^{\pi/2} \sec x\,dx$

ii). $\int_0^3 \frac{dx}{x-1}$.